

SCREENSHOTS OF PROJECT EXECUTION AND OUTPUTS

QUERY 1 -> SALES ROUTING & RAG RETRIEVAL (PRICING PLANS)

```
PS C:\Users\Akhilan\customer-support-assistant> python main.py

=====
Initializing system...
=====
[RAG] Vector store built with 20 chunks from docs/

#####
QUERY 1: What are the pricing plans available for your software?
Customer: Akilan (C001)
#####
[Intent Classifier] Query classified as: sales (requires_approval=False)
[RAG Retrieved] Source: pricing_guide.txt → PRICING GUIDE - ABC Technologies

SUBSCRIPTION PLANS

Starter Plan - $29/month
- ...
[RAG Retrieved] Source: pricing_guide.txt → Enterprise Plan - $199/month
- Unlimited team members
- 1TB cloud storage
- Adva...
[RAG Retrieved] Source: pricing_guide.txt → ADD-ONS
- Additional storage: $10/month per 50GB
- Additional team members beyon...
[Sales Agent] Response generated.
[Supervisor Agent] Final response validated and saved to memory.
```

```
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FINAL RESPONSE TO CUSTOMER:
Here is the revised response:

Dear [Customer],

Thank you for considering our pricing options at ABC Technologies. We have three main plans designed to meet the needs of different businesses: Starter, Professional, and Enterprise.

Our Starter Plan is ideal for small teams, offering up to 5 team members, 10GB cloud storage, basic reporting, email support, and core business management features for $29/month.

For growing businesses, our Professional Plan provides more advanced features, including up to 25 team members, 100GB cloud storage, priority email and chat support, API access, and custom workflows for $79/month.

Our top-tier plan is the Enterprise Plan, which offers unlimited team members, 1TB cloud storage, advanced reporting, analytics, and forecasting, as well as 24/7 phone, chat, and email support, full API access with higher rate limits, custom integrations, a dedicated account manager, and an SLA guarantee of 99.9% uptime for $199/month.

We also offer annual billing discounts of 20% on all plans for customers who choose to pay upfront for the year. If you have any questions or would like to discuss your specific needs, please don't hesitate to contact us.

Best regards,
[Your Name]
ABC Technologies Customer Support
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```

QUERY 2 -> ACCOUNT ROUTING & RAG RETRIEVAL (APPLICATION CRASH)

```
#####
QUERY 2: I forgot my account password.
Customer: David (C002)
#####
[Intent Classifier] Query classified as: account (requires_approval=False)
[RAG Retrieved] Source: technical_manual.txt → COMMON LOGIN ISSUES
If a customer cannot log in, check the following in order:
1...
[RAG Retrieved] Source: faq_document.txt → FAQ DOCUMENT - ABC Technologies

Q: How do I reset my password?
A: Click "Forgot..."
[RAG Retrieved] Source: faq_document.txt → Q: How do I deactivate my account temporarily?
A: Go to Account Settings > Account...
[Account Agent] Response generated.
[Supervisor Agent] Final response validated and saved to memory.

-----
FINAL RESPONSE TO CUSTOMER:
Final response:
To reset your password, please click on the "Forgot Password" link on our login page and follow the prompts. You will be asked to enter your registered email address, and we'll send you a temporary password reset link via email that is valid for 30 minutes. Please check your inbox for this link and follow its instructions to create a new password.
-----
```

QUERY 3 -> TECHNICAL SUPPORT ROUTING & RAG RETRIEVAL (AGENT ROUTING & RAG)

```
#####
QUERY 3: My application crashes whenever I upload a file.
Customer: Priya (C003)
#####
[Intent Classifier] Query classified as: technical (requires_approval=False)
[RAG Retrieved] Source: technical_manual.txt → APPLICATION CRASH TROUBLESHOOTING
If the application crashes during file upload:...
[RAG Retrieved] Source: technical_manual.txt → 6. If the issue persists after these steps, collect the crash log (Settings > Diagnostics)
[RAG Retrieved] Source: faq_document.txt → Q: What happens if my payment fails?
A: We will retry the payment 3 times over 7 days...
[Technical Agent] Response generated.
[Supervisor Agent] Final response validated and saved to memory.
```

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FINAL RESPONSE TO CUSTOMER:
Here's the final response:

"I'm here to assist you with troubleshooting the issue. Based on our technical manual, let's try the following steps to resolve the problem:

1. Check the file size: Ensure your uploaded file is within the allowed limit for your plan (500MB per file on Starter/Professional plans, 2GB on Enterprise).
2. Verify the file format: Confirm that the file type you're uploading is supported by our application (CSV, XLSX, PDF, DOCX, PNG, JPG).
3. Check your internet connection: Try uploading the file again with a stable internet connection.
4. Clear the application cache and restart: This can often resolve issues related to corrupted data or temporary glitches.

If none of these steps resolve the issue, you may want to try updating to the latest version of our application, as crash bugs are frequently patched in newer versions.

If the problem persists after trying these steps, I'd be happy to guide you through collecting and escalating a crash log for further investigation. Would you like me to walk you through that process?"
-----
```

QUERY 4 -> BILLING, RAG RETRIEVAL & HUMAN-IN-THE-LOOP APPROVAL (REFUND REQUEST)

```
#####
QUERY 4: I need a refund for my annual subscription.
Customer: Akilan (C001)
#####
[Intent Classifier] Query classified as: billing (requires_approval=True)
[RAG Retrieved] Source: company_policy.txt -> COMPANY POLICY DOCUMENT - ABC Technologies

REFUND POLICY
Customers are eligible...
[RAG Retrieved] Source: company_policy.txt -> CANCELLATION POLICY
Customers may cancel their subscription at any time from the...
[RAG Retrieved] Source: faq_document.txt -> Q: Do you offer a mobile app?
A: Yes, our mobile apps are available for iOS and ...
[Billing Agent] Response generated.

!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
HUMAN-IN-THE-LOOP APPROVAL REQUIRED
Customer query: I need a refund for my annual subscription.
Proposed response: To process a refund, I'll need to review your account and determine if you're eligible based on our Refund Policy.
```

```
Since you have an annual subscription, we offer prorated refunds for unused months after the 14-day window. Can you please provide me with your order date or subscription start date so I can check how many months are left in your current billing cycle?
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
Supervisor decision - approve this response? (y/n): y
[Human Approval] Approved by supervisor.
[Supervisor Agent] Final response validated and saved to memory.

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FINAL RESPONSE TO CUSTOMER:
Here's a revised version of the response:

"Thank you for reaching out about processing a refund. To assist you, I'll need to review our Refund Policy and determine if you're eligible for a prorated refund based on your annual subscription.

To proceed, could you please provide me with your order date or subscription start date? This will help me accurately calculate the remaining months in your current billing cycle."
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```

QUERY 5-> MEMORY STORAGE & RECALL (PREVIOUS SUPPORT ISSUE)

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#####
QUERY 5: What was my previous support issue?
Customer: Akilan (C001)
#####
[Intent Classifier] Query classified as: memory_recall
[Memory Recall] Answered using stored conversation history.
[Supervisor Agent] Final response validated and saved to memory.

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FINAL RESPONSE TO CUSTOMER:
Here's the revised response:

Based on our conversation history, it appears that previous support issues were related to pricing and billing. Specifically:

1. You initially inquired about our software pricing plans.
2. Later, you requested a refund for your annual subscription.

If you're experiencing an issue with either of these topics or have a new concern, please let me know how I can assist you today.
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PS C:\Users\Akhilan\customer-support-assistant>
```